Engineering Mechanics 1 · Unit_03 · UNIT 3

DRAFT — UE26ME142A · Unit_03 · UNIT 3 · CLASS 36 Centroid Numerical

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

CLASS 36 Centroid Numerical

1 CLASS 36 Centroid Numerical

Study focus: CLASS 36 Centroid_ Numerical. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000042 · EV0000043

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In today's class let's solve few problems in centroid.

Key details

- This is the first question in today's class.
- Problem 5/47.
- Determine the coordinates of the centroid of the trapezoidal area as shown in the figure.
- We have been asked to locate the centroid of this lamina.
- So what we can do here is we can use the method of approximation and we can locate the centroid of this lamina.
- To use the method of approximation, we know the formula x bar is equals to summation of EI Xi by.
- Total area y by summation of a i yi divided by total area.
- To use that formula, we have to divide this into elemental areas.
- How to divide this?
- Just draw a vertical line passing through this point.
- We are going to get one rectangle and one triangle.
- Together we will get a trapezium.
- The first one is rectangle.
- And what will.
- The width of the rectangle is 14 and the depth of the rectangle is 60.
- Then the second elemental area is.
- triangle here base of the triangle will be total 60 minus 40.
- After that the height of the triangle will be the height of the rectangle that is 60 ml.
- Next, we know the formulas x bar and y bar.

- Remember, since we have your two elemental areas, the formula is going to be a1 x1 plus a2 x2 divided by total area that is a1 plus a2.
- it by total area.
- That's what we have to find it out to find the value of x bar and y bar.
- A1 is nothing but area of the first elemental area.
- For time being, forget about this triangle.
- That will be equal to B into H.
- What is B from here?
- What is H?
- Here you will get it as 2400 mm square.
- Next, we need X1 and Y1.
- What is X1 and Y1?
- If you look at this rectangle, again it is a lamina.
- Each and every lamina will have a centroid.
- That will be this centroid.
- C1 stands for the centroid of this rectangular lamina.
- The coordinates of the C1 are X1, Y1.
- X1 is X coordinate of C1, Y1 is Y coordinate of the C1.
- Then what is this x1?
- We know for a rectangle from here the centroidal distance will be b by 2.
- Here b is 40.
- What is y1?
- From ear to ear distance that will be total will be sixty, sixty by two will be y one, that is h by two.
- You will get it as 30 mm.
- So we are done with the first elemental area.
- Next come to the second one.
- A2 will be the area of this triangle.
- What is A2?
- Half into base into height.
- Base is 20 and the height is 60.
- Substitute here, you will get the value of triangle A2.
- After that, we should know the value of X2 and Y2.
- X2 and Y2 are nothing but the coordinates of the centroid of the second lamina.
- Since this is a triangle here it will be having a centroid that is C2.
- The x coordinate of C2 is X2, y coordinate of C2 is Y2.
- Then how to find X2?
- Remember the triangular formula.
- From this point to here, it will be B by 3.
- But what is X2?
- X2 is from the y axis.
- That is X2.
- What is base here?
- Base is nothing but base of the triangle, that is 20.

- Substitute here, and we will get the value of X2 as 46.67.
- Next, we should know the value of Y2.
- We know the standard formula with respect to base of the triangle, the y coordinate of the centroid will be h by 3.
- H is 60.
- The formula is going to be like this: substitute the values of a1, x1, a2, x2 in this, and you'll get the value of x bar.

Example or application from the lecture

- This is the given lamina.
- Let's know about the dimensions of this rectangle.
- Now let's find out what is a1, what is x1, what is y1, next a2, x2, y2.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000042 · EV0000043 · EV0000044 · +1 source units in pack.json

Core explanation · 05:00–09:59

Substitute the values of a1, y one, a2, y2, you'll get the value of y bar.

Key details

- You'll get test twenty-eight point.
- The coordinates a centroid.
- The coordinates of the centroid of the trapezoidal area are 25.33 comma 28.
- That is what the meaning.
- Next question 5 slash 47.
- Determine X and Y coordinates of the center of the shaded area.
- We have been asked to locate the centroid of this lamina.
- And look at this, the y-axis is here at a distance of 50 with respect to this member of a rectangle.
- Again, we don't have a standard formula for this figure.
- We have to use the method of approximation.
- To use the method of approximation, we have to divide this into elemental areas.
- That's up to you.
- Here I have taken it as three elemental areas.
- One is this rectangle, one more is this rectangle, and one more is a triangle.
- The first one is a rectangle, here width of the rectangle is 50, and the depth of the rectangle is 200.
- Second one is again a rectangle of width 150 and the height or a depth 200.
- Next third one is a triangle of base 100 and the height 200.
- Now we know the formula in method of approximation.
- X bar is summation of AI Xi by total area, y bar is summation of AI yi divided by total area.
- Since here we have three elemental areas, the formula is going to be a one x one plus a two x two plus a three x three by total area that is a one plus a two plus a three.

- A1 Y1 plus A2 Y2 plus A3 Y3 divided by total area that is A1 plus A2 plus A3.
- First A1 is nothing but AF the first.
- Element since it is in the form of rectangle, area of this rectangle will be B into H.
- Next, what is X1 and Y1?
- This lamina will have a centroid.
- For time being, forget about this second rectangle and this triangle.
- This rectangle is the area so it will have a centroid C here.
- Some are here C one we will call it as.
- Then the X and Y coordinates of this centroid are nothing but X1, Y1.
- What is X1?
- The distance of this C1 with respect to y axis and y1 is the distance of the c1 with respect to x axis.
- Where is the c1?
- It is in the second quadrant.
- Will be positive.
- That's the reason.
- And x will be equals to b by 2.
- We know the value of b 50 from here, substitute here, 50 by 2 will be equals to 25.
- Since it is in second quadrant, x coordinate will be negative.
- And y1 will be h by 2.
- That will be hundred.
- Then come to the second one.
- Area of this will be equals to B into H.
- Here B is one fifty and H is two hundred.
- Substitute that you will get the value of A2.
- Okay, now for time being forget about first elemental area and the third one, this triangle.

Example or application from the lecture

- This is the given lamina area.
- Let's divide this into elemental areas as rectangle and triangle.
- Yeah, what you can do here is either you can go for only one rectangle of base 200 and you can consider this is one rectangle of 50 and one more rectangle of 150 and one triangle.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000046 · EV0000047 · EV0000048 · +1 source units in pack.json

Core explanation · 10:00–14:59

Just consider about this rectangle.

Key details

- This rectangle is again an elemental area, so it will have a centroid that is C2.
- X2, Y2 are the coordinates of this C2.
- Then what is X2?
- X2 will be B by 2.

- B we know 150.
- And Y2, we know that is Y2 is equal to H by 2.
- Here H is 200.
- We are done with the first two areas.
- A3 stands for area of the triangle that will be half into base into height.
- Base is 100 and the height is 200.
- Substitute you will get the value of A3.
- Again, it's a lamina, it will have a centroid that will be C3.
- The coordinates of this C3 will be X3 from year to ear distance and Y3 will be this distance.
- Then how to find this X3 year?
- We know from the standard formulas the distance.
- Central with respect to this line will be B by 3.
- Please remember the formulas.
- But X3 is measured with respect to y-axis.
- From year to year it is 150 plus b by 3.
- And y three we know for a triangle the y the distance of the centroid with respect to its base will be h by three.
- Plus into minus you will get this entire term as negative, whereas the second term is positive, third term is also positive.
- Just simplify this, you will get the value of x1.
- Then y1 is.
- If you simplify you'll get the value of y1 as 93.34 mm.
- The coordinates of this android are 76.67 millimeter and 93.34 millimeter.
- That's the point where we assume the entire area of the lamina is concentrated.
- This is the next question 5 slash 51.
- Determine X and Y coordinates of the centroid of the shaded area.
- What is shaded here?
- It's not like a trapezium.
- In this trapezium, the circle has been removed.
- Then how to locate the center end of only this remaining area.
- Again, we don't have a formula, standard formula, so we have to follow the method of approximation.
- To follow the method of approximation, we have to divide this into elemental areas.
- Think how to get this shaded area.
- Earlier in one of the previous problem we just divide the trapezium into rectangle and triangle.
- But here if we divide this as triangle and rectangle, will we get this shaded area?
- Then in that what do we need to detect?
- We need to detect this circle.
- I am going to draw a horizontal line passing through this point for Tm.
- Then only we will get this shaded area.
- Will it be fifteen?
- No, because here it is 40.
- This is 40, these two are paralleline, so this height will be 40.

- Then what will be the height of the triangle?
- Total minus 40.
- From ear to ear it is 80.
- 80 minus 40 will.
- Give us 40 mm.
- Don't take it as 50.
- Here the first one is a rectangle of width 100 mm and height 40 mm.
- Second one is this triangle, base of the triangle will be 100 and the height of the triangle will be 40.

Example or application from the lecture

- Now let's start with method of approximation.
- Now let's start with the first one.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000050 · EV0000051 · EV0000052 · +1 source units in pack.json

Core explanation · 15:00–19:59

Third one is deduction, remember that.

Key details

- And it's the circle of radius.
- The formula remains the same summation of AI xi by total area, summation of AI divided by total area.
- A1 stands for area of the first elemental.
- Since it's the rectangle, area will be b into h.
- Next, we need to find the value of x1 y1.
- Since it's the lamina, it will have a centroid that will be C1.
- C1 stands for centroid of this rectangle.
- Then x1, y1 are the x and y coordinates of this c1.
- Then what is x1?
- It will be since it's a rectangle, we have the y axis here, so it will be equal to b by 2 and y1 will be equal to h by 2.
- Next, we have to start with the second area.
- Here the second one is a triangle.
- Next, we know the value of base and we know the height of the triangle.
- Substitute here you will get the value of area of the triangle.
- When we talk about.
- The triangle we should forget about other areas.
- This triangle is again a laminary area, it will have a centroid that is C2.
- X2, Y2 are nothing but X and Y coordinates of this C2.
- Then what is this X2?
- We know in a for triangle the formula this x coordinate with respect to this side will be B by 3.

- Then what is y2?
- Y2 is this distance total.
- But in the standard formula, we know this distance, that is the distance of the centroid with respect to its basis H by 3.
- That is from this to this is h by 3.
- But what is y2 total.
- Then the last one is circle.
- Since area we have to deduct here, area doesn't exist here, it should be taken as negative.
- As being removed from this trapezium, so area itself is negative.
- area is minus pi r square, you will get less 1256.63 mm square.
- It will have a centroid that will be C3, then the coordinates of that will be X3, Y3.
- The sign of X and Y coordinates of the centroid depends upon where is the centroid, in which quadrant the centroid is.
- Since the centroid is in first quadrant, both X3 and Y3 will be positive.
- Then what is Y3 here?
- Just substitute the formulas.
- The formula is going to be a1x1 plus a2x2 plus a3x3 by a1 plus a2 plus a3.
- That is the general formula.
- I don't think you will get any doubts in first two terms.
- Once you simplify you take the value of x bar as 45.62 mm.
- substitute all the values from this tablet column and then.
- You'll get the value of y bar.
- Y bar is thirty one point three seven.
- That's the point where we assume the entire area is concentrated.
- Expir by bar.
- That's about today's class.
- If you come across any doubt, you can post your queries in Pesum forum.

Example or application from the lecture

- Now let's start with the method of approximation.
- Let's start with the first elemental area.
- Just consider this rectangle.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000054 · EV0000055 · EV0000056 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
CLASS 36 Centroid_ Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/035_CLASS_36_Centroid__Numerical__8d2e6c9175d9d883.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 36 Centroid_ Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/035_CLASS_36_Centroid__Numerical__8d2e6c9175d9d883.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 36 Centroid_ Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/035_CLASS_36_Centroid__Numerical__8d2e6c9175d9d883.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
CLASS 36 Centroid_ Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/035_CLASS_36_Centroid__Numerical__8d2e6c9175d9d883.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 15 rendered pages in 033_Centroid__9f0e55503970de0e.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.slides.md have no readable `.slides.md` text; their original page images are included.
3. 3 rendered pages in 039_UE24CV141A__Unit_3_Beams__9fc34bbd5ca57988.slides.md have no readable `.slides.md` text; their original page images are included.
4. 3 rendered pages in 039_UE25CV141A_Unit_3_Class_48_Lecture_PPT__4d28f283532e284e.slides.md have no readable `.slides.md` text; their original page images are included.

5. 3 rendered pages in 042_UE25CV141A_Unit_3_Class_53_Lecture_PPT__c20af5d61d14aaf0.slides.md have no readable `.slides.md` text; their original page images are included.
6. 5 rendered pages in 043_UE25CV141A_Unit_3_Class_54_Lecture_PPT__c7db1f404daea47d.slides.md have no readable `.slides.md` text; their original page images are included.
7. 6 rendered pages in 044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.slides.md have no readable `.slides.md` text; their original page images are included.
8. 4 rendered pages in 044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.slides.md have no readable `.slides.md` text; their original page images are included.
9. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
10. 6 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
11. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
12. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
13. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
14. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
15. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00010	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...
SRC00011	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00012	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...
SRC00013	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...

C Revision and retrieval practice